Implement your own ping program

Exp. no: 13

Date : 01/10/2026

Aim:

To implement a simple ping program in computer networks to check whether a destination host is reachable and to measure the round-trip time (RTT) of packets.

Algorithm :

→ Start the program
→ Read the destination host name or IP address from the user
→ Create a raw ICMP socket.
→ Construct an ICMP Echo Request packet containing :
1. ICMP Type
2. Code
3. checksum
4. Identifier
5. Sequence number
6. Data
→ calculate the checksum of the packet.
→ Send the ICMP Echo Request to the destination.
→ wait for an ICMP Echo Reply.
→ calculate the round-trip time between sending the request and receiving the reply.
→ Display the destination, packet size, sequence number and RTT.

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→ Repeat the process for a specified number of packets.

→ Display the final result and terminate.

program :- python

```
import socket
import struct
import time
import os
def checksum(data):
    total = 0
    length = len(data)
    for i in range(0, length-1, 2):
        total += (data[i] << 8) + data[i+1]
    if length % 2:
        total += data[-1] << 8
    total = (total >> 16) + (total & 0xffff)
    total += total >> 16
    return (~total) & 0xffff
def ping(host):
    try:
        dest_addr = socket.gethostbyname(host)
    except socket.gaierror:
        print("Unable to resolve host")
        return
    icmp = socket.getprotobyname("icmp")
    try:
        sock = socket.socket(socket.AF_INET, socket.SOCK_RAW, icmp)
```

```
except permissionError:
    print("permission denied. Run the program with
    administrator / root privileges.")
    return
sock.settimeout(2)
packet_id = os.getpid() & 0xFFFF
sequence = 1
# ICMP Echo Request
header = struct.pack("!BBHHH", 8, 0, 0, packet_id,
                     sequence)
data = b"computer Networks ping"
packet = header + data
# calculate checksum
chksum = checksum(packet)
# Re-create header with checksum
header = struct.pack("!BBHHH", 8, 0, chksum,
                     packet_id, sequence)
packet = header + data
start_time = time.time()
try:
    sock.sendto(packet, (dest_addr, 1))
    sock.recvfrom(1024)
    end_time = time.time()
    rtt = (end_time - start_time) * 1000
    print("Reply from", dest_addr, ": bytes =",
          len(packet), "time = %.2f ms" % rtt)
except socket.timeout:
    print("Request timed out")
```

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finally :

sock.close()

Main program

host = input ("Enter hostname or IP address:")

for i in range (4):

ping(host)

time.sleep(1)

Note : Raw ICMP sockets usually require administrator/root privileges, on Linux, run with sudo; on windows, run the terminal with appropriate administator privileges.

Output :

Enter hostname or IP address : 8.8.8.8

Reply from 8.8.8.8: bytes = 43 time = 24.51 ms

Reply from 8.8.8.8 : bytes = 43 time = 23.87 ms

Reply from 8.8.8.8 : bytes = 43 time = 24.12 ms

Reply from 8.8.8.8 : bytes = 43 time = 24.36 ms.

Result :

Thus, a ping program was successfully implemented using python. The program sends ICMP echo request packets to the specified host, receives Echo Reply packets and calculates the round-trip time (RTT) to determine network connectivity.

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